

Assignment 12

Zeynep Cakez

2024-11-16

Exercise 1

- i. How many images in total are in the Kuzushiji-MNIST dataset? (In the full dataset, not the reduced version we are using in this assignment.)

The dataset has 70,000 images in total

- ii. What are the dimensions (in pixels) of each image?

Each image is 28x28 pixels in grayscale.

- iii. How many labels are there?

There are 10 labels, matching the first 10 characters from the Japanese Hiragana script.

Exercise 2

```
train_dl <- train_ds %>%  
  dataloader(batch_size = 32, shuffle = TRUE)
```

Exercise 3

```
batch <- train_dl$.iter()$.next()  
batch[[1]]$size()
```

```
## [1] 32 784
```

- i. what do these two numbers mean? (I.e. why are they the values that they are?)

The two numbers are 32 and 784. 32 is the batch size, representing the number of images in one batch. And 784 is the number of pixels in each image, since each image is 28x28 in grayscale.

Exercise 4

```
visualize_image(5)
```



- i. Which of the 10 Japanese characters does your image correspond to? You should report the phonetic English version of the character (i.e. what it sounds like) instead of the Hiragana symbol.

The Japanese Character I think my image responds to is ma.

Exercise 5

```
valid_dl <- dataloader(
  valid_ds,
  batch_size = 32
)

test_dl <- dataloader(
  test_ds,
  batch_size = 32
)
```

Exercise 6

```
net <- nn_module(
  initialize = function() {
    self$fc1 <- nn_linear(28 * 28, 128)
    self$fc2 <- nn_linear(128, 10)
  },
  forward = function(x) {
    x <- x %>%
      self$fc1() %>%
      nnf_relu() %>%
      self$fc2()
    return(x)
  }
)
```

```
)
```

Exercise 7

```
model1 <- net %>%  
  setup(  
    loss = nn_cross_entropy_loss(),  
    optimizer = optim_adam,  
    metrics = list(  
      luz_metric_accuracy()  
    )  
  )  
)
```

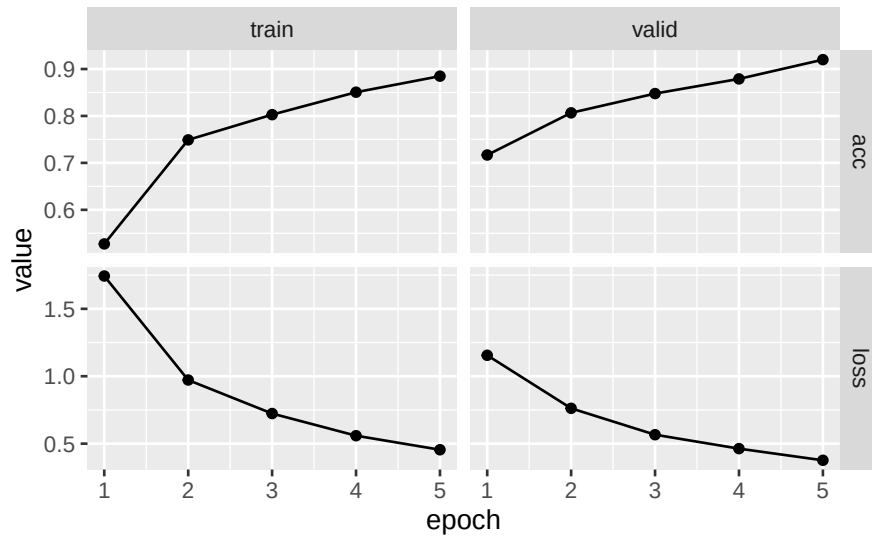
Exercise 8

```
acc <- accelerator(cpu=TRUE)  
fitted1 <- model1 %>%  
  fit(  
    train_dl,  
    epochs = 5,  
    valid_data = valid_dl,  
    verbose = TRUE,  
    accelerator = acc  
  )
```

```
## Epoch 1/5  
## Train metrics: Loss: 1.7432 - Acc: 0.5273  
## Valid metrics: Loss: 1.1553 - Acc: 0.7168  
## Epoch 2/5  
## Train metrics: Loss: 0.9717 - Acc: 0.749  
## Valid metrics: Loss: 0.7619 - Acc: 0.8066  
## Epoch 3/5  
## Train metrics: Loss: 0.7232 - Acc: 0.8027  
## Valid metrics: Loss: 0.5662 - Acc: 0.8477  
## Epoch 4/5  
## Train metrics: Loss: 0.5589 - Acc: 0.8506  
## Valid metrics: Loss: 0.4629 - Acc: 0.8789  
## Epoch 5/5  
## Train metrics: Loss: 0.4547 - Acc: 0.8848  
## Valid metrics: Loss: 0.3764 - Acc: 0.9199
```

Exercise 9

```
plot(fitted1)
```



i. Patterns in the graphs:

The training and validation accuracy graphs show positive growth, meaning there is an increase in accuracy over time. On the other hand, the training and validation loss graphs demonstrate negative growth, meaning that the loss decreases as the model improves its performance.

Exercise 10

```
test_dl <- dataloader(
  test_ds,
  batch_size = 32
)
evaluate(fitted1, test_dl, accelerator = acc)
```

```
## A `luz_module_evaluation`
## -- Results -----
## loss: 1.1836
## acc: 0.6484
```

i. How does this accuracy and loss on the test dataset compare to the training and validation sets (from Exercises 8 & 9)? Why is this?

The test dataset shows lower accuracy and higher loss compared to the training and validation sets because the model has become too specialized in the training data. This is known as overfitting, where the model learns the specifics of the training data too well and struggles to generalize to new, unseen data. It's common for models to perform worse on the test set since it contains data the model has not encountered before.

Academic Integrity statement

No tools other than the teachers guidance from Slack and the textbook chapter on Deep Learning was used to complete this assignment